

ADVITH KRISHNAN

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EDUCATION

B.Tech in Artificial Intelligence

SRM Institute of Science and Technology

Aug 2021 – Aug 2025

CGPA: 8.44/10.00

Relevant coursework: Neural Networks & Machine Learning, Deep Learning Techniques, Reinforcement Learning, Inferential Statistics, Image & Video Processing, Advanced Calculus.

PROFESSIONAL EXPERIENCE

Systems Engineer

Infosys

Dec 2025 – Present

Chennai, India (Onsite)

- Working with a Nasdaq-100 & Fortune-500 logistics client on large-scale data backends, facilitating a legacy EDW migration (Informatica, Teradata) of ~9 billion rows to Microsoft Azure.

Software Engineer

StejasSYS

Apr 2025 – Sept 2025

Atlanta, USA (Remote)

- Implemented proprietary end-to-end RAG, agentic, and computer-vision features for clients in the health industry.
- Deployed AI models on GCP; built CRUD flows to Supabase with Docker for containerization and Directus for CMS.
- Designed Next.js frontends & FastAPI interfaces; owned full-stack delivery across frontend, backend, and DB features.

Research Intern

ETH Zurich

Aug 2024 – Mar 2025

Zurich, Switzerland (Remote)

- Simulated motor-neuron behaviour to study PNS recruitment patterns at the Institute of Neuroinformatics and the [Neural Bionics Lab](#) (Chalmers University of Technology), using the NEURON and NEST frameworks.
- Programmed customizable nerve-fiber models to simulate 3 invasive neurological experiments; explored deployment of intraneural models on a mixed neuromorphic chip.

AI Engineer Intern

MIOT International

Dec 2023 – Jul 2024

Chennai, India (Onsite)

- Built 2 anchor few-shot models (PyTorch, Weights & Biases) for shoulder-bone loss estimation and dislocation-type identification, training on only 60 patient scans.
- Fine-tuned models to improve diagnostic consensus across 57 surgeons; lectured clinicians on the methods.

Research Intern

NASA

Sep 2022 – Oct 2023

Washington, D.C., USA (Remote)

- Redesigned the IMPACT team's satellite-image dataset (4 years of sparse multispectral footage of North America) via a 3-stage pipeline (arcGIS, Rasterio, Keras).
- Validated the pipeline with ResNet-34/50/104 and Inception-v3; ran EDA and statistical inference on satellite datasets.

OPEN-SOURCE EXPERIENCE

PyTorch | Open Source Contributor | [Merged PRs](#)

Jul 2026 – Present

- Implementing ATen operators for the Metal Performance Shaders (MPS) backend of `pytorch/pytorch`.
- Fixed a BLAS buffer-overflow bug affecting PyTorch matmul operations.

Linux Foundation | Open Source Mentee | [Proof of Selection](#)

Jun 2025 – Oct 2025

- Selected for the LFX Mentorship 2025 (under Open Mainframe Project) with a \$3000 stipend (~1% acceptance rate).
- Built a RAG framework connecting IBM Z / IMS mainframes and modern databases for unified retrieval, enabling root-cause and trend analysis on large-scale infrastructure; published findings (Oct 2025).

PUBLICATIONS

[1] A. Krishnan, V. R. Viswasadhas, et al. "RAG to Riches: Understanding Actionable Insights from Mainframe Data," Open Mainframe Project, Oct. 2025. [\[link\]](#)

[2] P. Gupta, A. Krishnan, et al. "ViDAS: Vision-based Danger Assessment and Scoring." In *Proc. ICVGIP 2024*, ACM, Article 29, 1–9. [\[link\]](#)

[3] A. L. Amutha, A. Krishnan, et al. "Optimizing Smoke Detection: A Hybrid U-Net Approach with Advanced CNN Models through Satellite Image Segmentation." (*Under review, TFSC*)

[4] A. Krishnan and S. Y. Sait. "Understanding Fine-grained Classification with Deep Learning through Fish Species Identification." (*Bachelor Thesis*)

PROJECTS

Deep Divergence Graph Kernel | TensorFlow, NetworkX, NumPy, SciPy | [Code](#) | [Paper](#)
Refactored TensorFlow 2.x implementation of Google Research’s DDGK algorithm for unsupervised protein classification.

Contrastive Learner | TensorFlow, PyTorch, Keras | [Code](#) | [Paper](#)
Implementation of a contrastive-learning DenseNet model for chest X-ray images.

NeuStyle | TensorFlow, PyTorch, Keras | [Code](#)
Neural style-transfer model with a VGG-16 preprocessing backbone.

TouchGrass | Convex, React, TypeScript, TanStack Query, LangChain | [Code](#)
LLM-powered web application for habit-building and personal development.

SKILLS

AI/ML	PyTorch, TensorFlow, Keras, LangChain, CrewAI, AutoGen, wandb, MLflow, OpenCV, ONNX, Ollama, Llama, NumPy, SciPy, Matplotlib, Pinecone.
Full-Stack	Next.js, React, TanStack Query, FastAPI, REST, Microservices, PostgreSQL, Supabase, Databricks, Snowflake, Datadog, Postman.
Deployment	AWS, GCP, Vercel, Docker, Kubernetes, Linux, Git.
Languages	Python, C, C++, Java, JavaScript/Node.js, TypeScript, SQL, Bash, HTML, CSS.

COMMUNITY & VOLUNTEERING

Board Member Next Tech Lab	Dec 2023 – Aug 2025 <i>Chennai, India (Onsite)</i>
• Drove long-term initiatives for this internationally recognized research lab (QS Silver Award for Reimagining Education). • Mentored and lectured 30+ members on medical image segmentation, LLMs, and spiking neural networks.	
Research Team Head & Founding Member Odyssey Lab	Dec 2023 – Mar 2025 <i>Chennai, India (Onsite)</i>
• Combined xAI methods (GradCAM, Integrated Gradients) to evaluate multimodal LLMs/VLMs (GPT-4 Turbo, GPT-4o, Gemini 1.5 Pro) on dangerous-scenario perception. • Built VQA and Temporal Action Localization pipelines for danger assessment; led 6 interns and presented at ICVGIP.	